



Inspiring Excellence

BRAC UNIVERSITY

CSE 350: Digital Electronics and Pulse techniques

Exp-02: Analysis of the binary weighted and R-2R D/A converter

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Objectives

1. To construct binary weighted and R-2R D/A converter
2. Verifying that the digital signal is converted to a proportional analog signal

Equipment and component list

Equipment

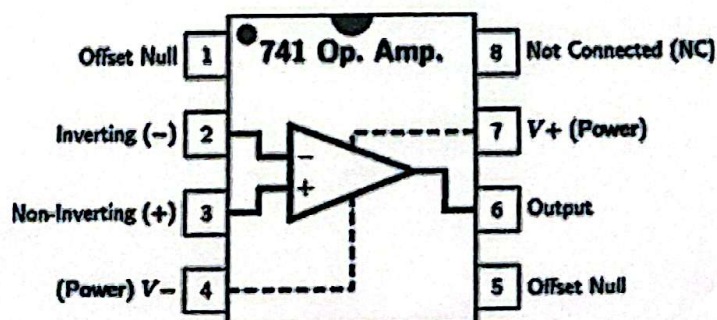
1. Digital Multimeter
2. DC power supply

Component

- Operational amplifier - UA741 - x1 piece
- Resistors -

- ❖ 10 K Ω - x1 piece
- ❖ 5 K Ω - x1 piece

- ❖ 2.5 K Ω - x1 piece
- ❖ 1.25 K Ω - x1 piece
- ❖ 1 K Ω - x1 piece



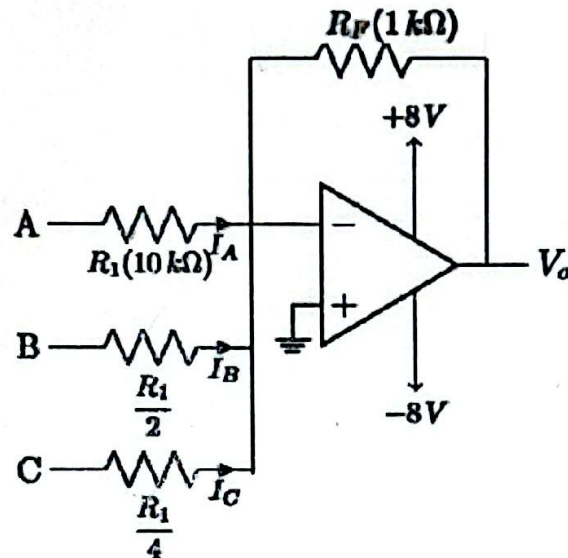


Figure 1: Binary Weighted D/A Converter

Task-01: Binary weighted D/A converter

THEORY

A three bit converter will have $2^3 = 8$ input combinations. Consequently, the converter will show 8 different output analog voltage levels for 8 different input combinations.

Case 1: (C, B, A) = (0, 0, 1)

The voltage across R_1 is 5V. So, the current through R_1 is $I_A = 0.5$ mA. Since the current into the op-amp input terminals are negligible, this 0.5 mA current will flow through the R_F resistance. Hence, the voltage across the resistance R_F is, $V_{RF} = 0.5 \text{ mA} \times 1 \text{ K}\Omega = 0.5 \text{ V}$. Consequently, the output voltage is -0.5V.

Case 2: (C, B, A) = (0, 1, 0)

The voltage across $R_1/2$ is 5V. So, the current through $R_1/2$ is $I_B = 1$ mA. Since the current into the op-amp input terminals are negligible, this 1 mA current will flow through the R_F resistance. Hence, the voltage across the resistance R_F is, $V_{RF} = 1 \text{ mA} \times 1 \text{ K}\Omega = 1\text{V}$. Consequently, the output voltage is -1V.

Case 3: (C, B, A) = (0, 1, 1)

The voltage across R_1 is 5V and the voltage across $R_1/2$ is 5V. The current through R_1 is $I_A = 0.5$ mA and the current through $R_1/2$ is $I_B = 1$ mA. So, the total current through the resistance is 1.5 mA. Hence, the voltage across the resistance R_F is, $V_{RF} = 1.5 \text{ mA} \times 1 \text{ K}\Omega = 1.5\text{V}$. Consequently, the output voltage is -1.5V.

Similarly, we can calculate the output voltage for any other input combination. The relationship between the digital input values and the corresponding analog output levels is presented in the staircase plot in figure 2.

The output is a negative going staircase waveform with 7 steps of -0.5V each. In practice, due to the variations in the logic HIGH voltage levels, all the steps will not have the same size. The value of the feedback resistor R_F changes the size of the steps. Thus, a desired size for a step can be obtained by connecting an appropriate feedback resistor. The only condition to look out for is that the maximum and minimum output voltages should not go beyond the saturation levels of the op-amp. We can find that the output voltage is defined by the expression:

$$V_O = \left(\frac{V_A}{R_1} + \frac{V_B \times 2}{R_1} + \frac{V_C \times 4}{R_1} \right) \times (-R_F) \quad (1)$$

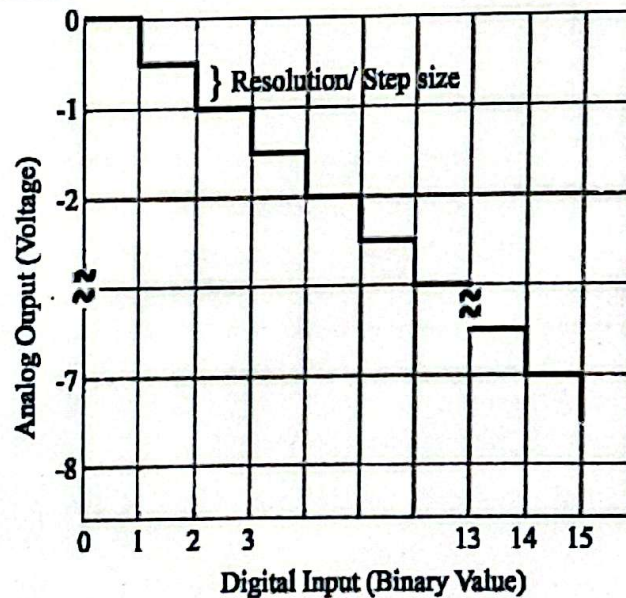


Figure 2: Staircase plot of output vs input in binary weighted D/A converter.

The x-axis of the plot represents the digital input values expressed in binary form. The y-axis represents the analog output voltage levels generated by the DAC in response to these digital inputs. The staircase appearance of the plot reflects the quantization process inherent in DAC operation. Each step corresponds to a specific digital input value, resulting in a discrete change in the analog output. This quantization means that the output can only take on certain values. The height of each step in the staircase plot indicates the resolution or step size of the DAC. A higher resolution DAC will have smaller steps, allowing for finer adjustments in the analog output. Conversely, a DAC with lower resolution will have larger steps, resulting in a more abrupt change in output for each increment in digital input. To find the resolution, one just needs to find the change in output analog voltage for one LSB change in digital input. As this DAC circuit is fundamentally an inverting summing amplifier, the output is negative, and decreases as we increase the digital input. This is the result of the inverting nature of this configuration.

Task-02: R/2R Ladder D/A converter

THEORY

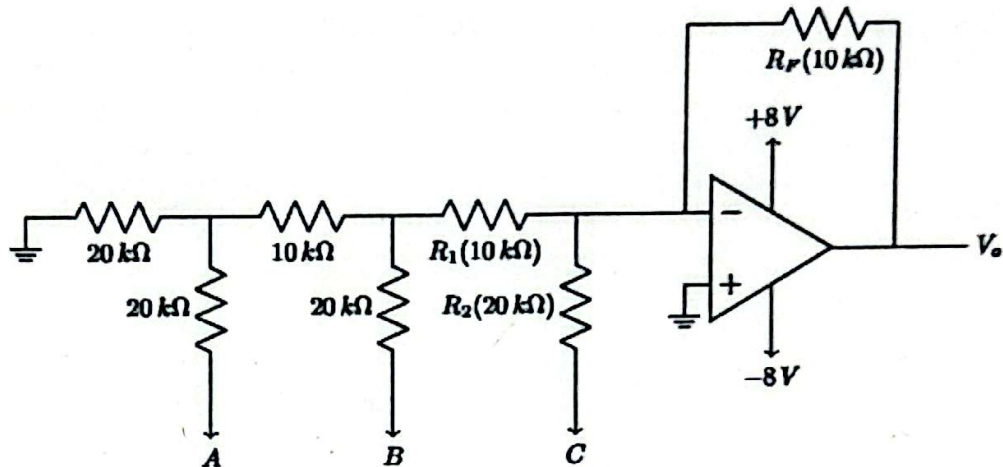


Figure 3: R2R Ladder D/A Converter

A digital to analog converter with R and 2R resistors is shown in figure 2.

As in the binary- weighted resistors converter, the binary inputs are simulated by the switches A-C and the output is proportional to the binary inputs. Binary inputs can be either in the HIGH (+5V) or LOW (0V) state.

The circuit can be solved using thevenin theorem. We will obtain an output vs input plot similar to that in figure 3 for this converter as well. We can find that the output voltage is defined by the expression:

$$V_O = \frac{1}{4} (V_A + 2V_B + 4V_C) \times \left(-\frac{R_F}{R_2}\right) \quad (2)$$

Procedure:

1. Construct the circuits on a breadboard. Supply +8V and -8V to the op amp.
2. Consider the **HIGH** input to be 5V and the **LOW** input to be 0V.
3. Use a multimeter to measure the output voltage for different input combinations. This output voltage is the 'analog' output signal. Fill up tables 1 and 2.

Precautions:

1. For the digital inputs, do not use the switches from the trainer board. The DAC might not work properly due to the current limit of the switches. You can use the 5V source of the trainer board or use DC power supply. If you are using multiple power supplies (trainer board and DC power supply simultaneously), make sure to connect the grounds of each individual supply.
2. For the op amp supply voltage, use the +8V and -8V from the trainer board (top left). Make sure to adjust the knobs to get the desired voltages.
3. Use a minimal number of wires. Unnecessary wiring increases the probability of errors.

4. Build the circuit similar to the layout of the given circuit diagram, such that debugging is easier if anything goes wrong.

Data Table 1

In all the data tables, write the input combinations in ascending order.

SL	V_C (V)	V_B (V)	V_A (V)	V_Y (V)
0	0	0	0	-0.0004
1	0	0	5.02	-0.50
2	0	5.02	0	-1.07
3	0	5.02	5.02	-1.57
4	5.02	0	0	-1.82
5	5.02	0	5.02	-2.32
6	5.02	5.02	0	-2.89
7	5.02	5.02	5.02	-3.39

Table 1: Table for binary-weighted D/A converter

Lab Task-01:

- a. Find the resolution of the Binary Weighted D/A converter.

Binary Weighted DAC	0.5
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- b. For the Binary Weighted D/A converter, change the value of R_F (feedback resistance) to $0.5 \times R_F$ and then to $2 \times R_F$. For each case, measure output voltage for any two consecutive input combinations (e.g. 001, 010) (V_{o1} & V_{o2}) and calculate the step sizes. Does the effect on step size match with the theory?

	Binary Weighted DAC		
	V_{o1}	V_{o2}	$\Delta = V_{o1} - V_{o2}$
$0.5 \times R_F$	-0.272	-0.58	0.308
$2 \times R_F$	-1.11	-2.34	1.23

001 010

In theory, the output voltage is directly proportional to the feedback resistor, R_F . So, if R_F is reduced to $0.5 \times R_F$, the step size should become half. If R_F is increased to $2 \times R_F$, the step size should become double. Now, $1.23 \approx 4 \times 0.308$, which makes sense because going from $0.5 \times R_F$ to $2 \times R_F$ is a 4 times increase in resistance. Since output step size is proportional to R_F , the step size also increases by about 4 times. Therefore, the effect on step size matches with the theory.

Data Table 2

SL	V_C (V)	V_B (V)	V_A (V)	V_Y (V)
0	0	0	0	-0.002
1	0	0	5.02	-0.62
2	0	5.02	0	-1.25
3	0	5.02	5.02	-1.87
4	5.02	0	0	-2.51
5	5.02	0	5.02	-3.12
6	5.02	5.02	0	-3.73
7	5.02	5.02	5.02	-4.38

Table 2: Table for R/2R ladder D/A converter

Lab Task-02:

- a. Find the resolution of the R/2R ladder D/A converter.

R/2R ladder DAC	0.63
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- b. For the R/2R ladder D/A converter, change the value of R_F (feedback resistance) to $0.5 \times R_F$ and then to $2 \times R_F$. For each case, measure output voltage for any two consecutive input combinations (e.g. 001, 010) (V_{o1} & V_{o2}) and calculate the step sizes. Does the effect on step size match with the theory?

	R/2R ladder DAC		
	V_{o1}	V_{o2}	$\Delta = V_{o1} - V_{o2}$
$0.5 \times R_F$	-0.295	-0.60	0.305
$2 \times R_F$	-1.23	-2.47	1.24

001 010

Yes, the effect on step size matches with the theory for the R/2R ladder D/A converter approximately.

The output voltage is directly proportional to R_F . So if R_F is reduced to $0.5 \times R_F$, the step size should become half. If R_F is increased to $2 \times R_F$, the step size should become double.

Here, $1.24 \text{ V} \approx 4 \times 0.305 \text{ V}$, which makes sense because going from $0.5 \times R_F$ to $2 \times R_F$ implies that the resistance increased by a factor of 4, and the step size also increased by about 4 times.

Signature

Signature

Report

Please answer the following questions briefly in the given space.

1. How can you get output lower than -15 V in the above D/A converter?

Ans. To get an output lower than -15V, we would need to increase the magnitude of the output voltage. This can be done by increasing the value of the feedback resistor R_F or by increasing the reference /input voltage. However, the output voltage cannot go beyond the negative saturation limit of the op-amp. So, to actually obtain less than -15V, we would also need to use an op-amp with higher supply voltages, because the output is limited by the op-amp's power supply.

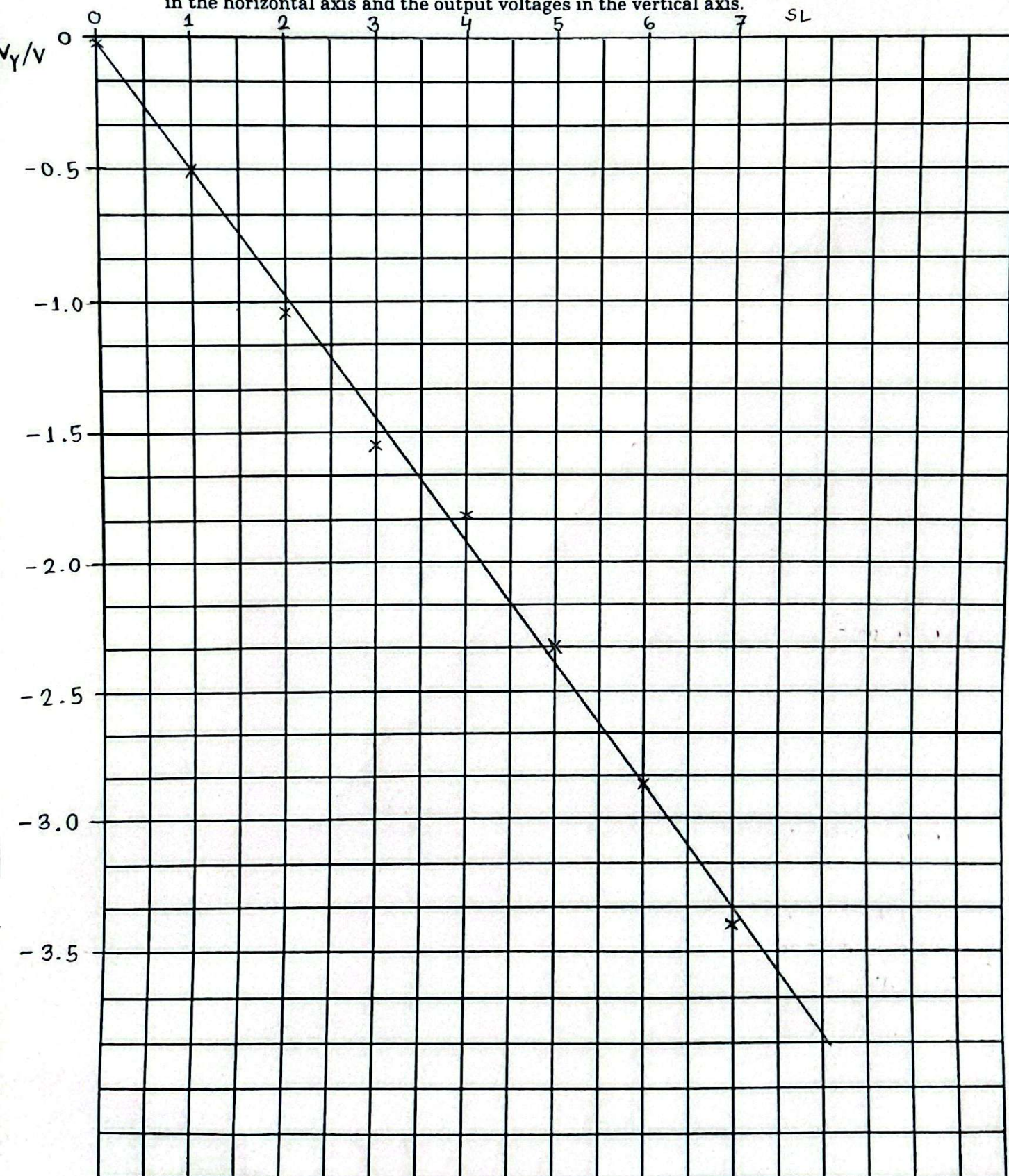
2. Briefly discuss which of the two converters is better in a practical scenario.

Ans. In a practical scenario, the $R/2R$ ladder DAC is better than the binary-weighted DAC. This is because the $R/2R$ ladder uses only two resistor values (R and $2R$), which makes it easier to design. In contrast, the binary-weighted DAC requires many different resistor values with precise ratios, which becomes difficult and less practical for higher bit numbers.

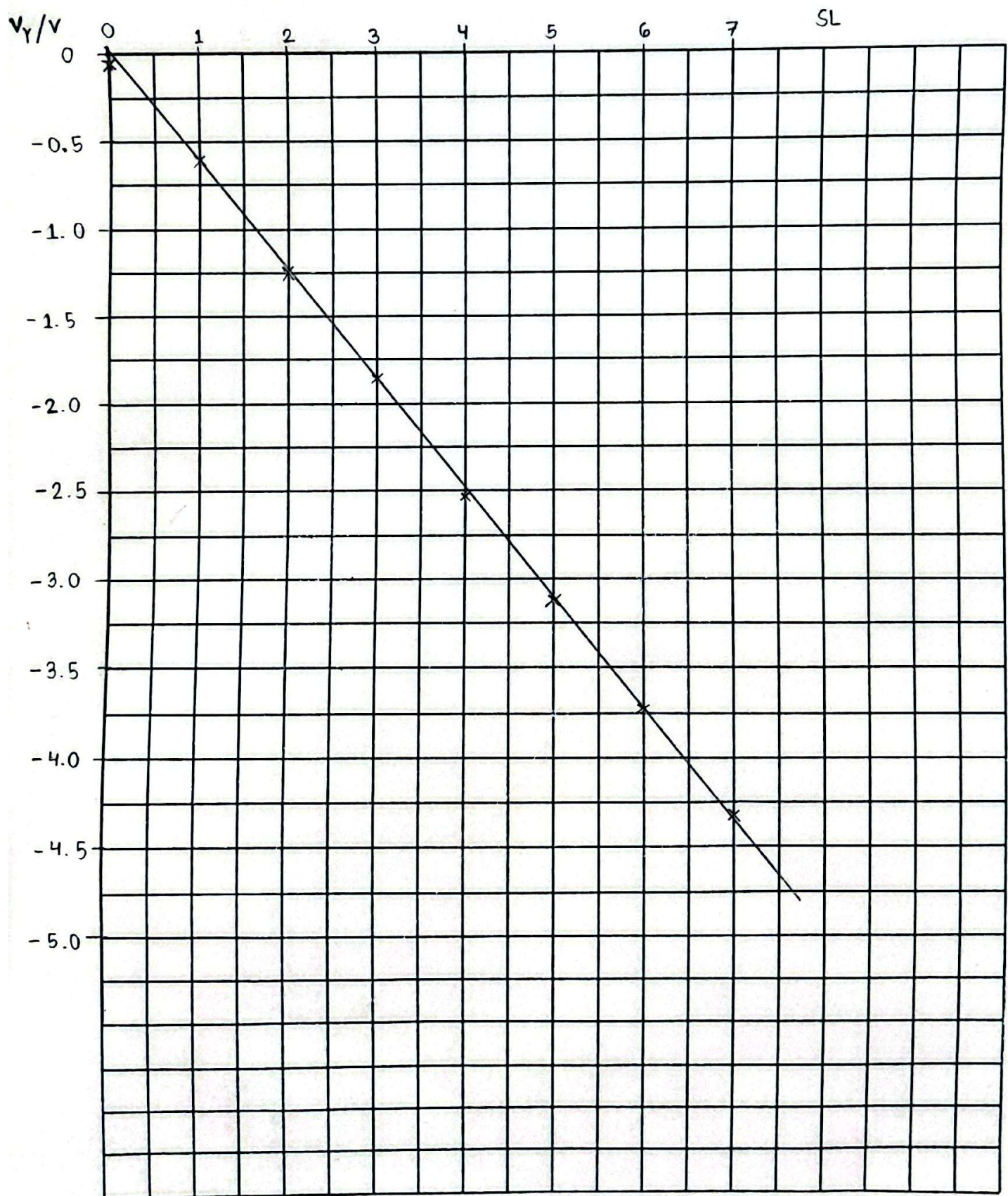
3. Write a discussion and include the following: your overall experience, accuracy of the measured data, difficulties experienced, and your thoughts on those.

Ans. Overall, the experiment was interesting and helped us understand how DAC circuits work in real life. Most of our measured values were close to the theoretical values, but at first we were getting wrong outputs because we accidentally used a wrong resistor in the circuit. After checking the connections and replacing it with the correct resistor, the output values became much closer to what we expected.

4. Plot the results obtained in table 1 and 2 in the given graph papers. Keep the serial no of inputs in the horizontal axis and the output voltages in the vertical axis.



Data Table 1



Data Table 2